Multithreading Programming (intro)

Taller de Programacion

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How can you 3 (or more) apps at the ***same time*** but with ***only*** 2 CPUs?

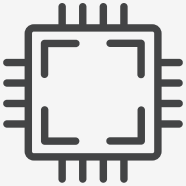
Pidgin
(chat client)

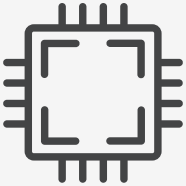
Rhythmbox
(music player)

Firefox
(web browser)

How the UI can process user's actions while still processing + reproducing the audio ?

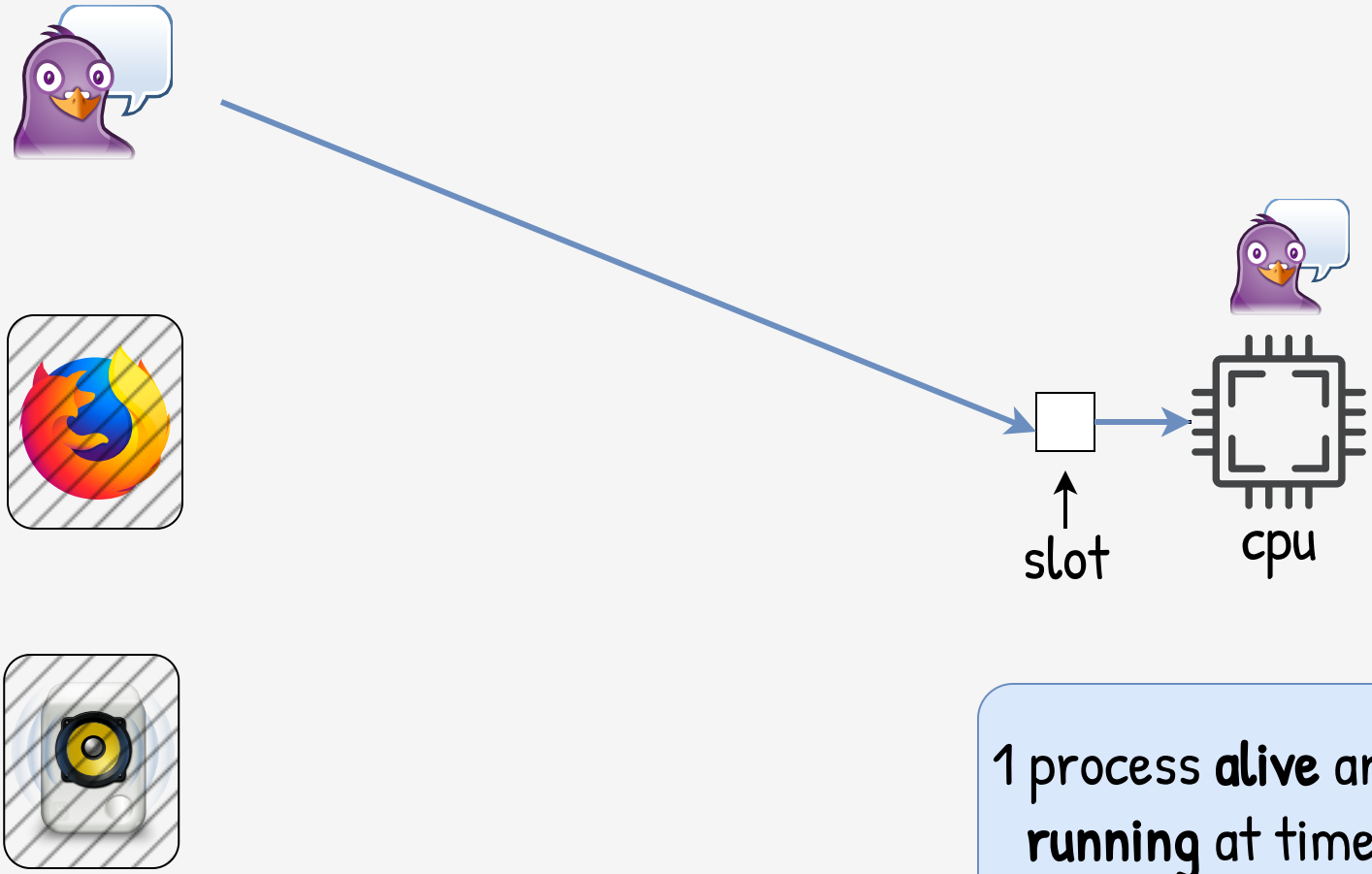
How the Firefox does load and run multiple pages at the same time?





running apps

CPUs cores



1 process **alive** and **running** at time

```
void process() {
    show_interface();
    read_messages_from_others();
    format_messages();
    show_messages();
    wait_local_message();
    show_local_message();
    send_local_message()
}
```

```
void process() {
    show_interface();
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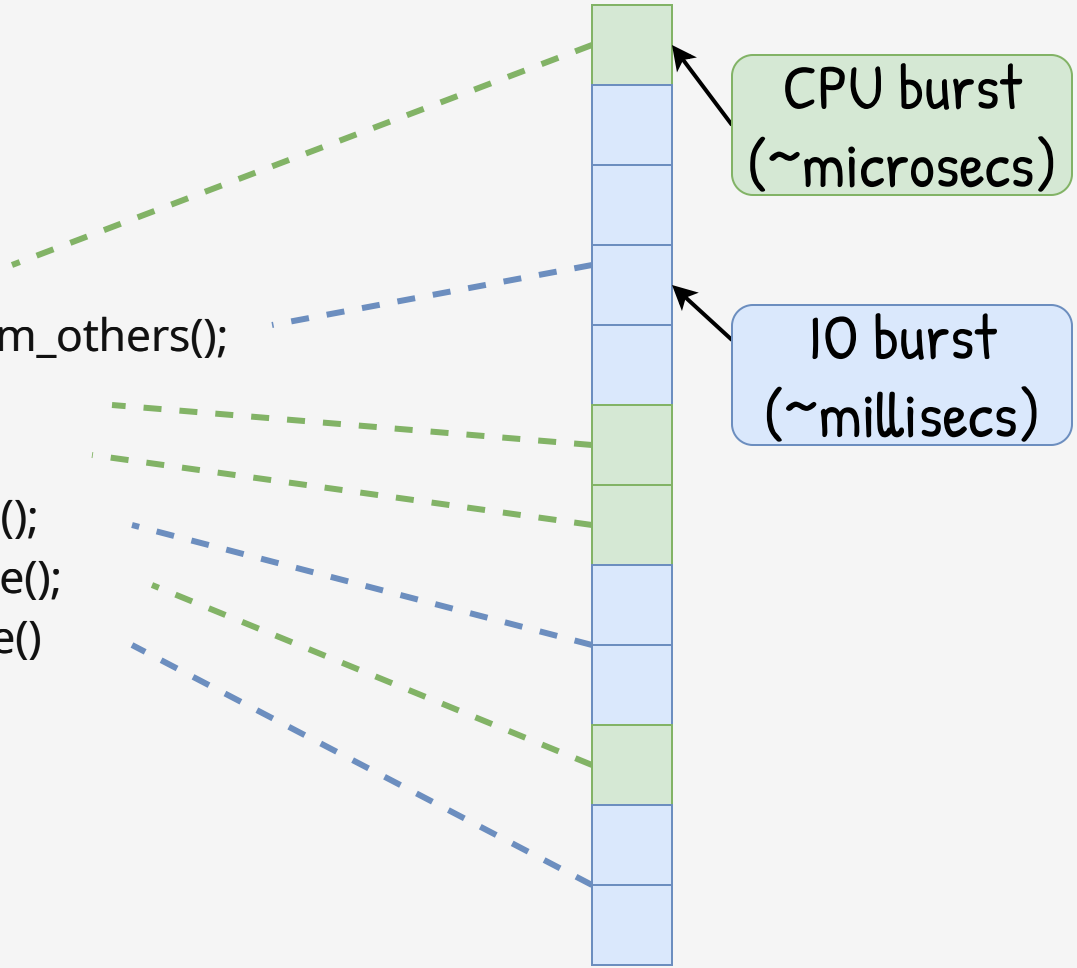
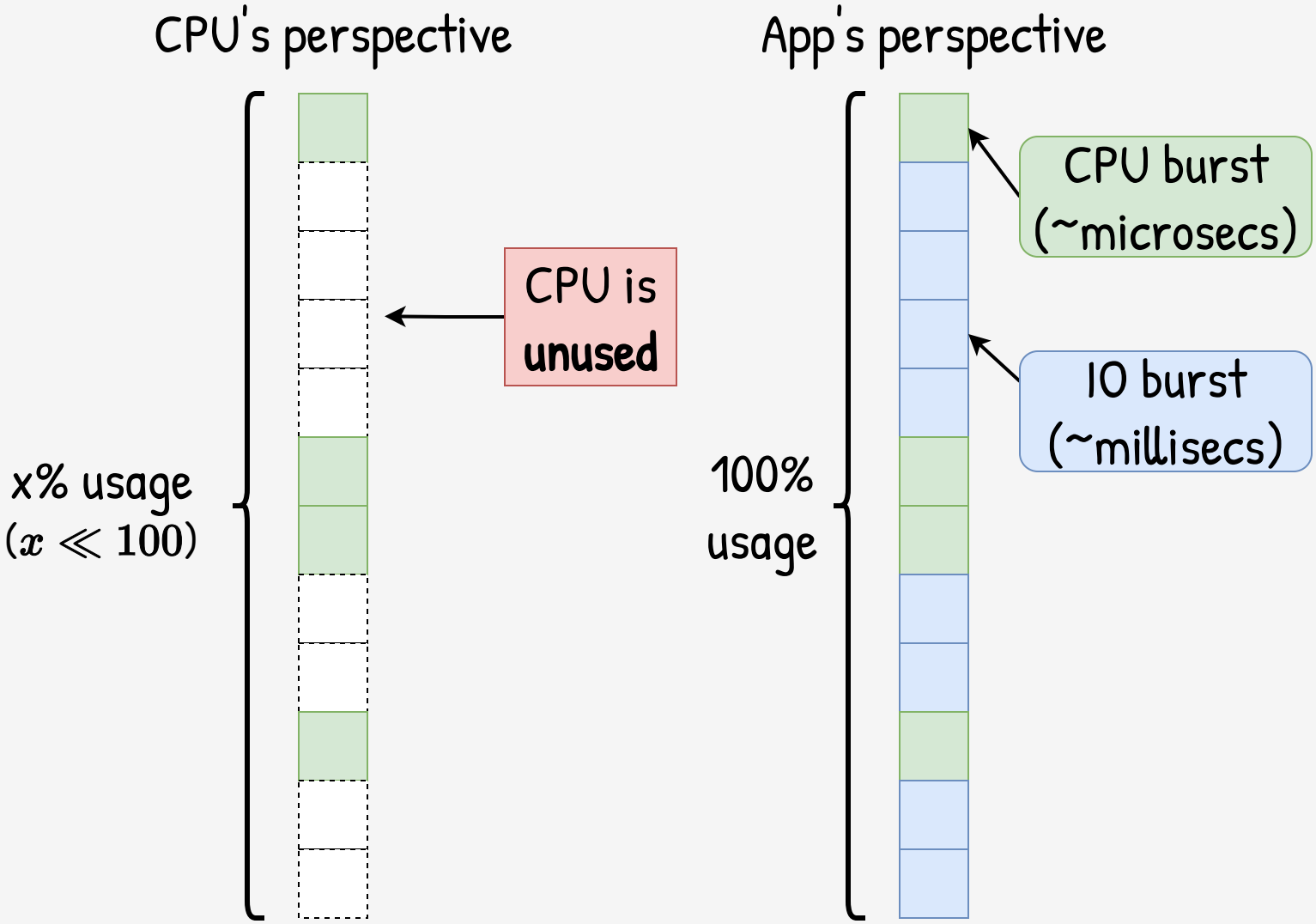
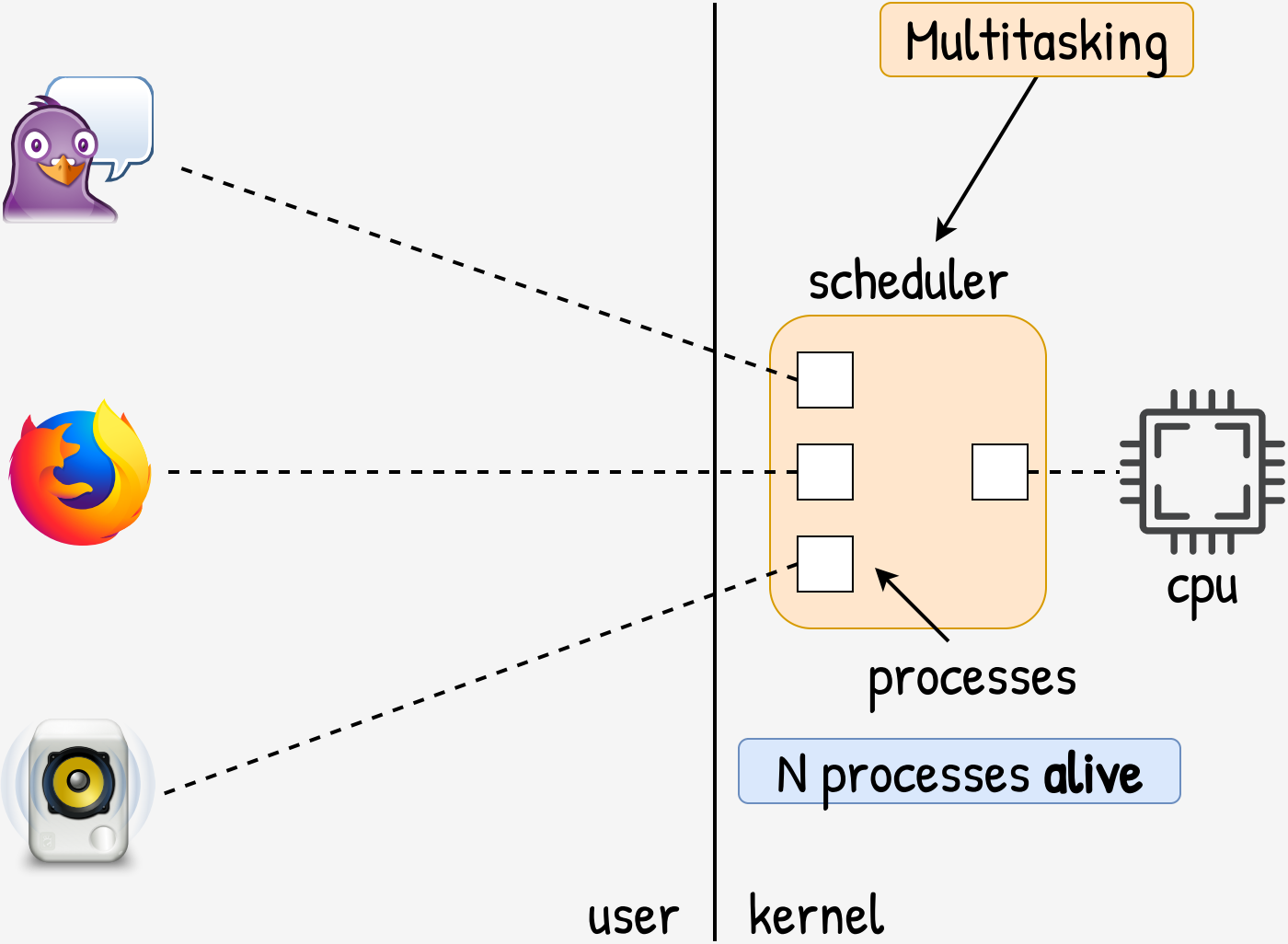
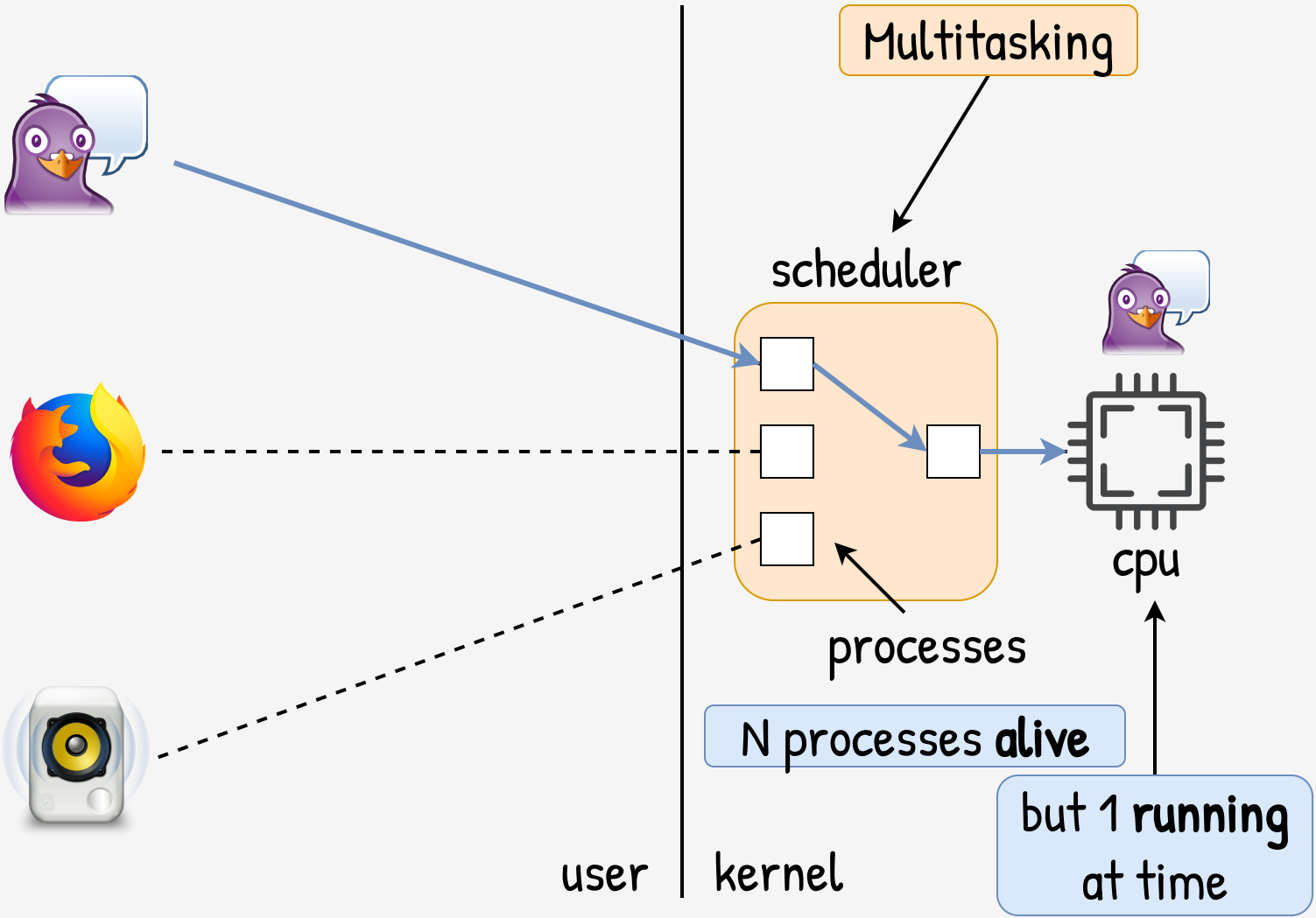



CPU's perspective
App's perspective
CPU is unused
x% usage
$(x \ll 100)$
100% usage
CPU burst (~microsecs)
IO burst (~millisecs)


Multitasking
scheduler
cpu
processes
N processes alive
user
kernel


Multitasking
scheduler
cpu
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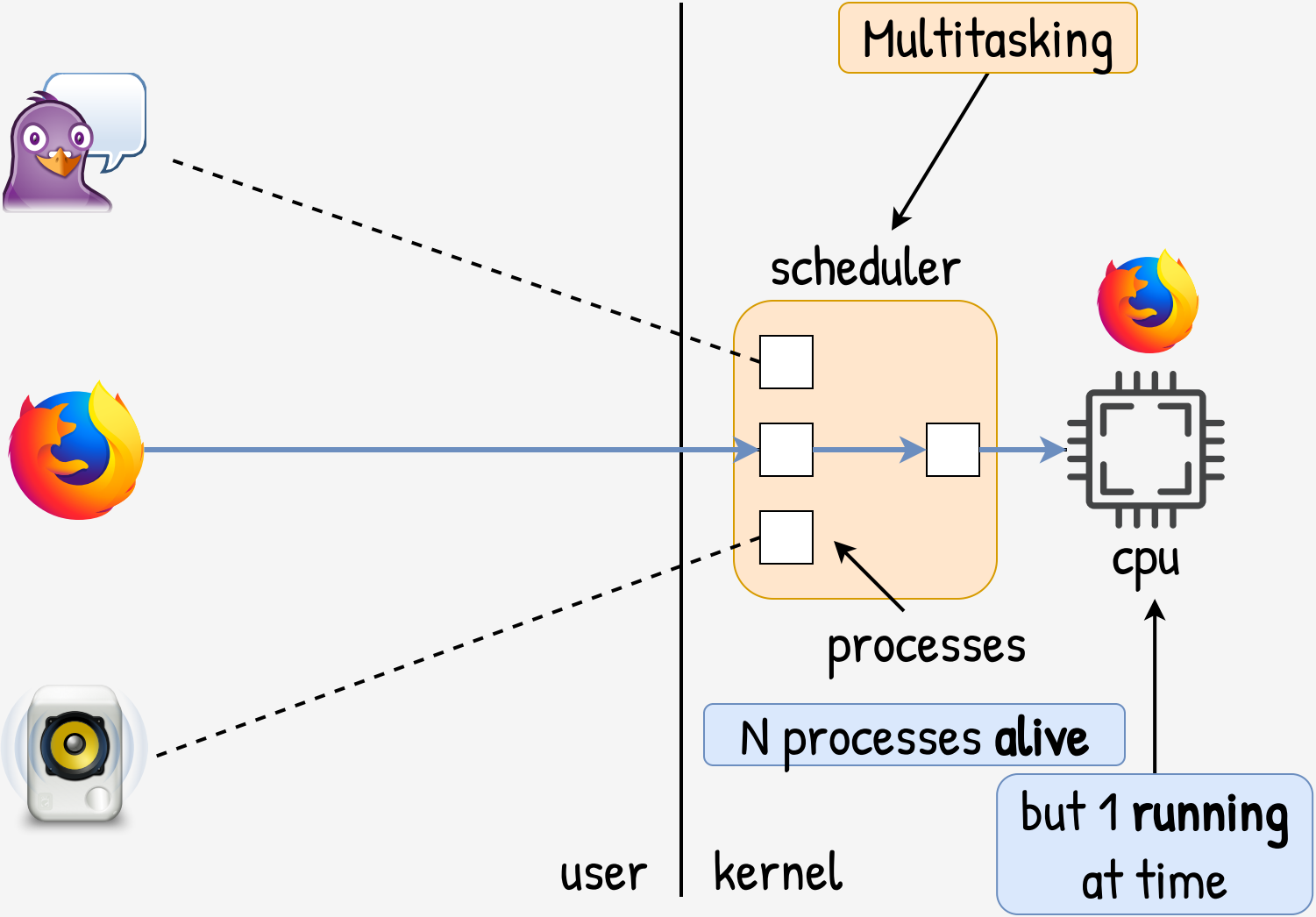
N processes **alive**
but 1 **running** at time
user
kernel


Multitasking
scheduler
processes
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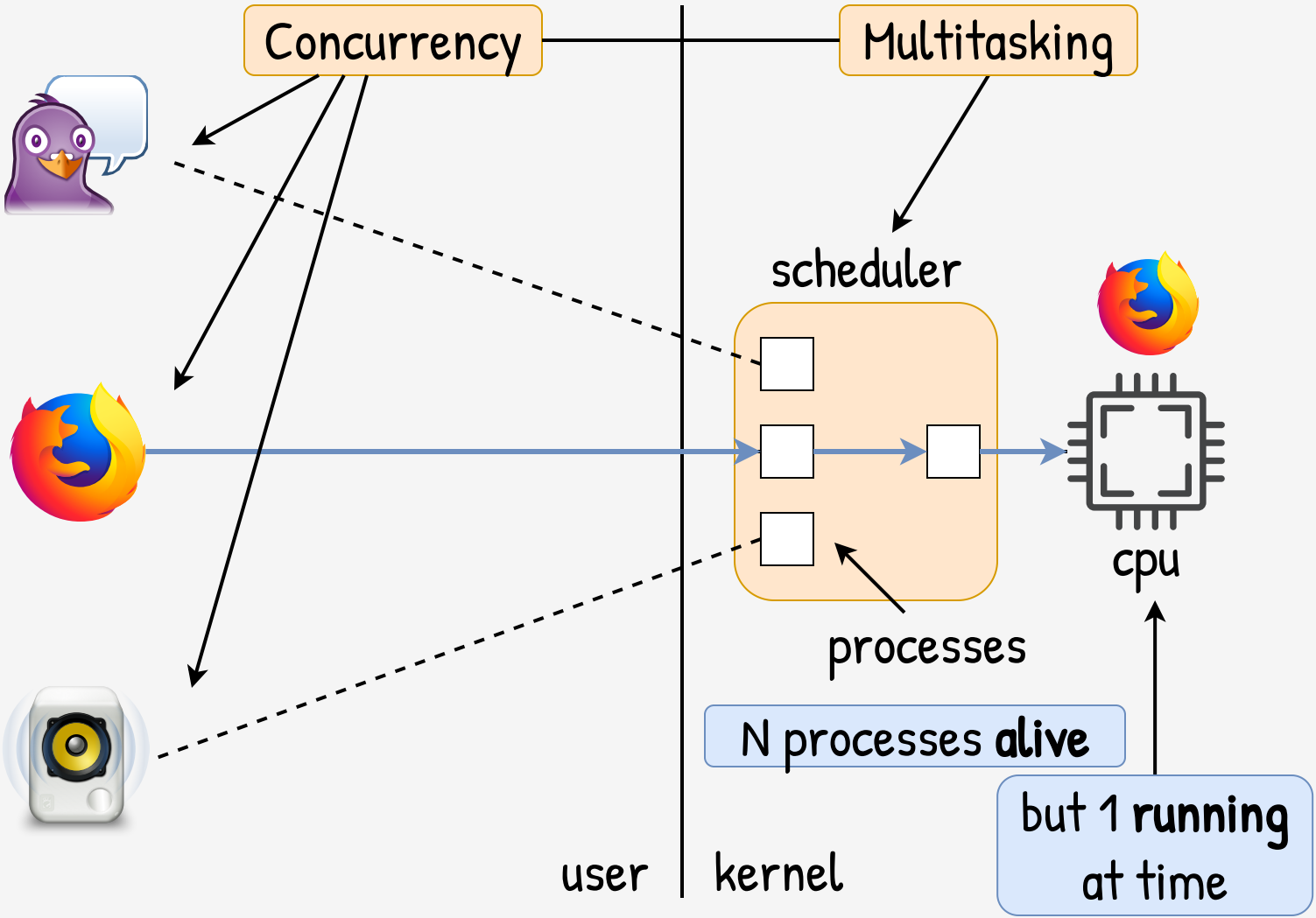
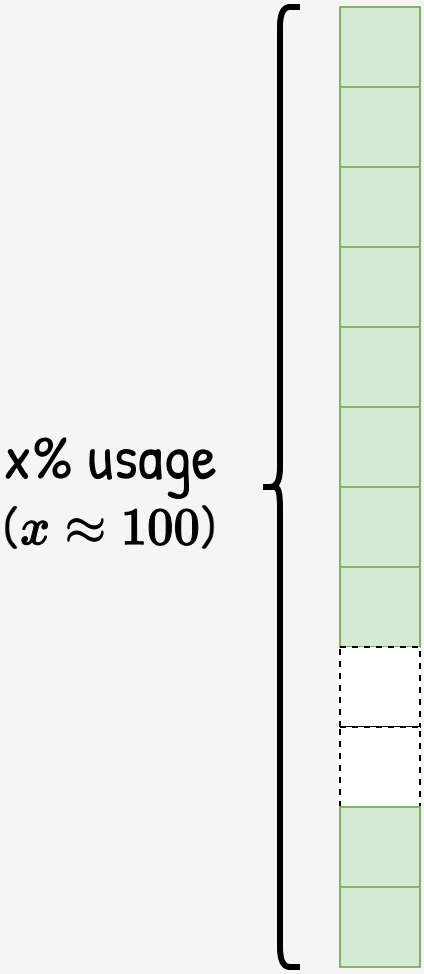
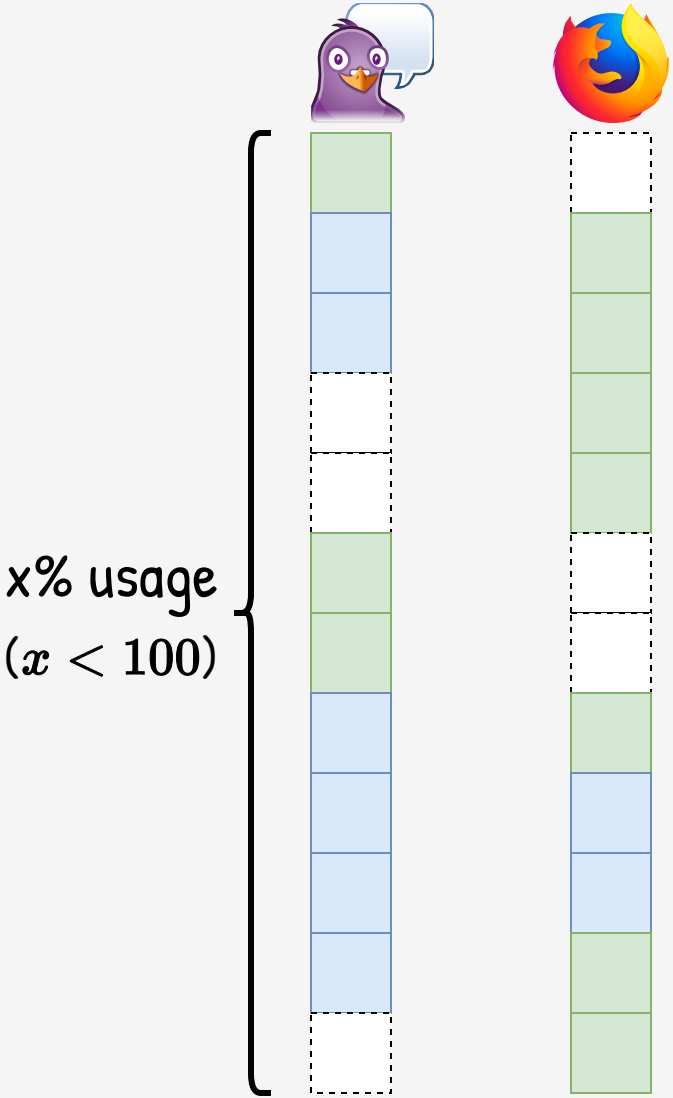
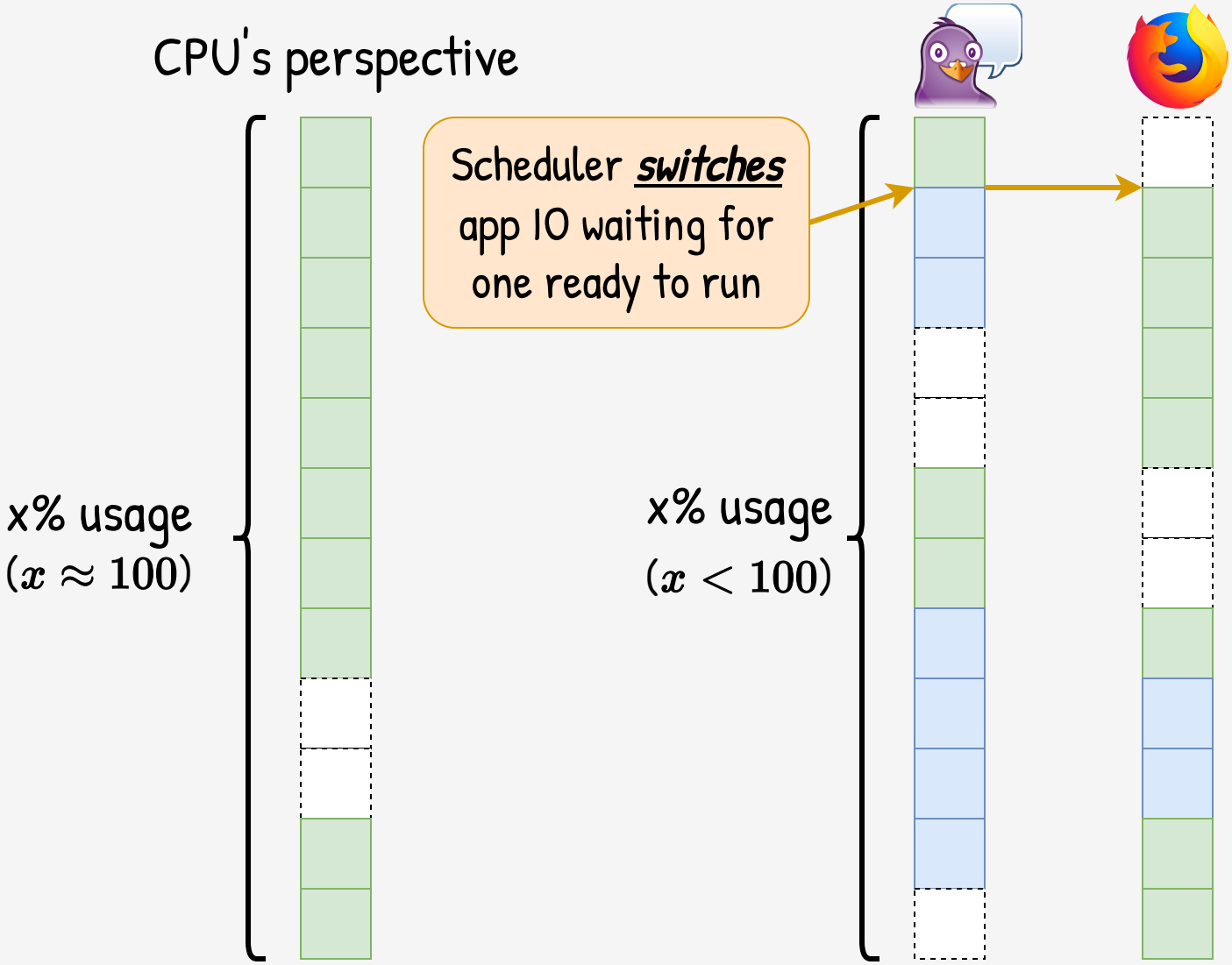
N processes **alive**
but 1 **running** at time
user
kernel

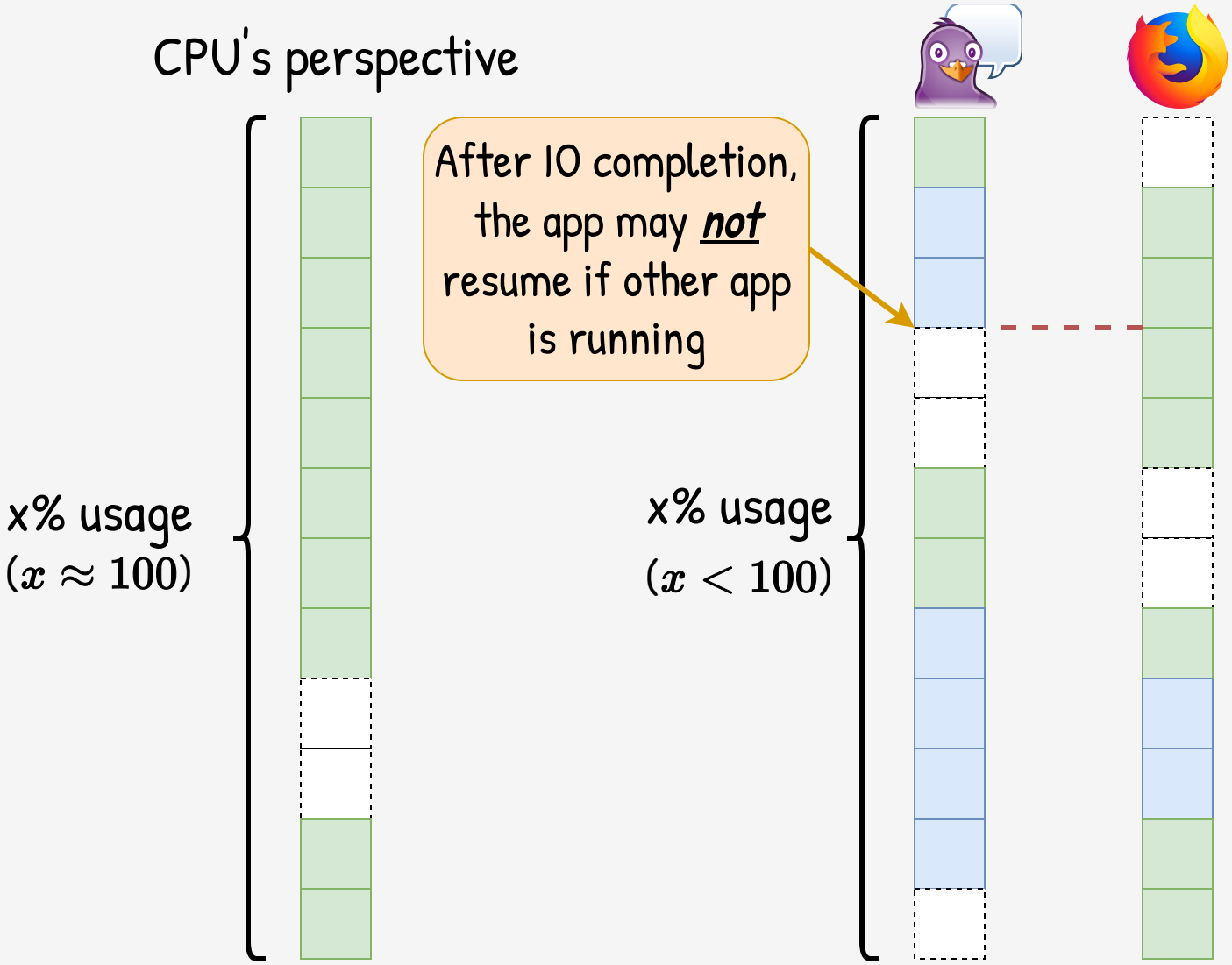

Concurrency
Multitasking
scheduler
cpu
processes
N processes **alive**
but 1 **running** at time
user
kernel

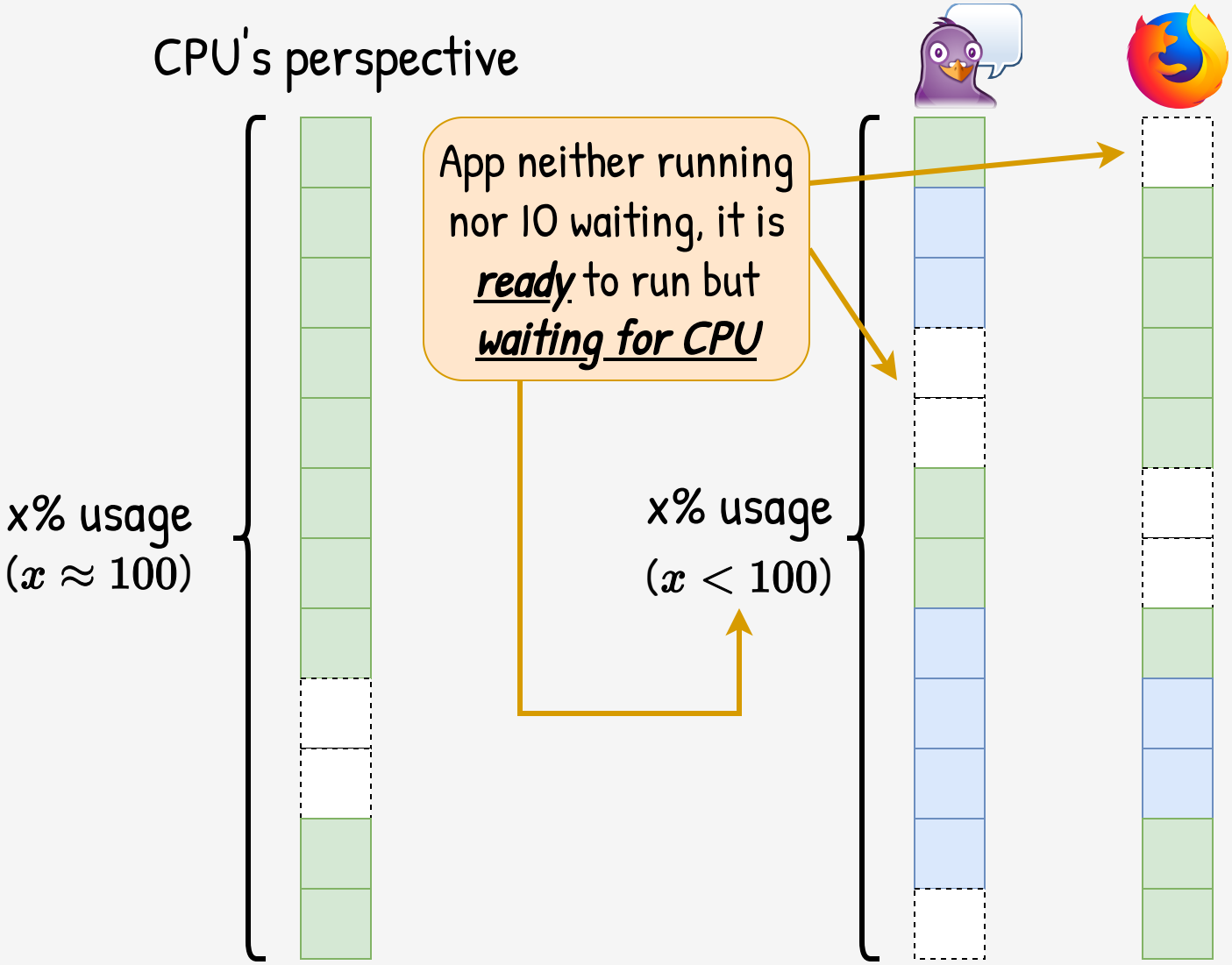
CPU's perspective

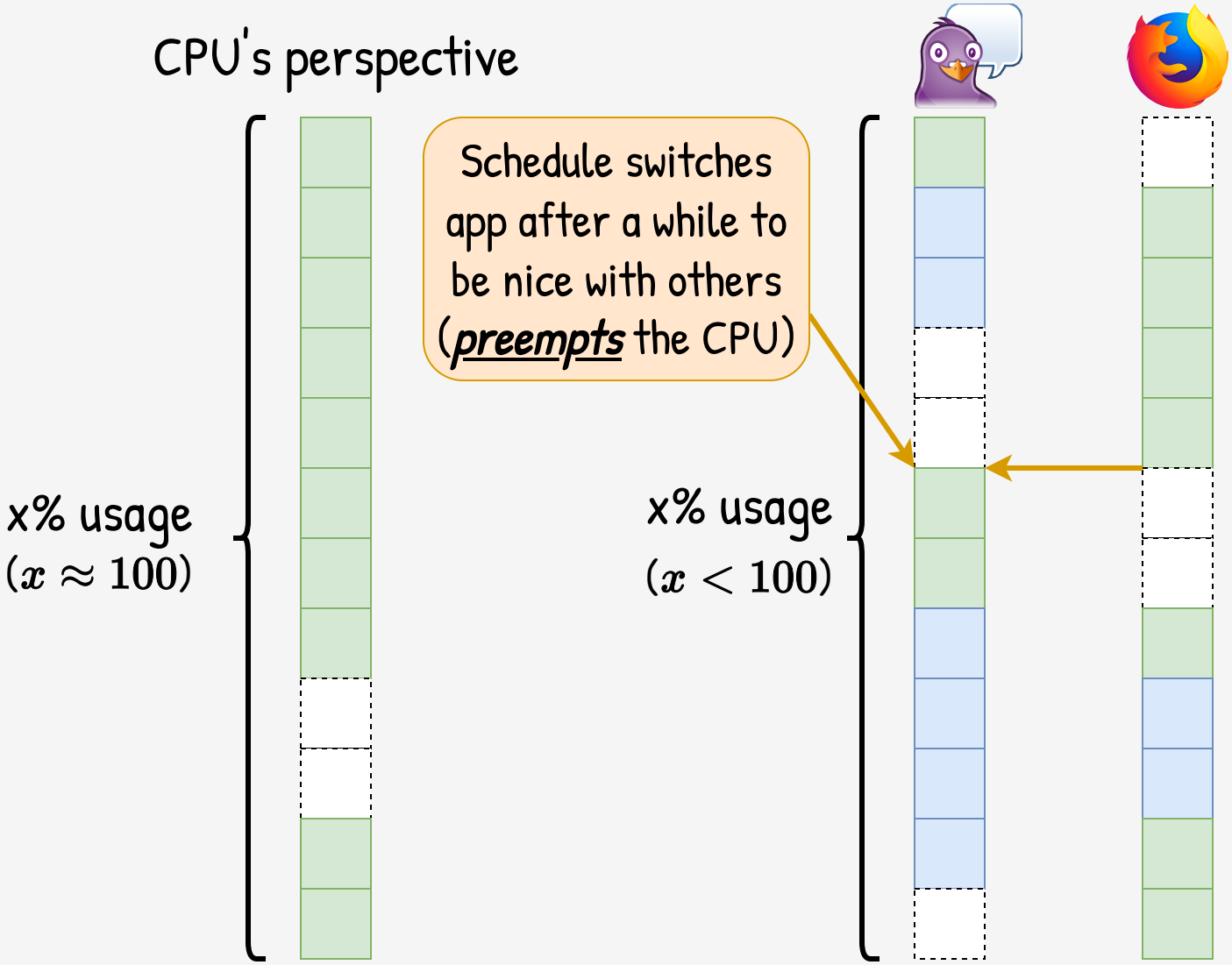


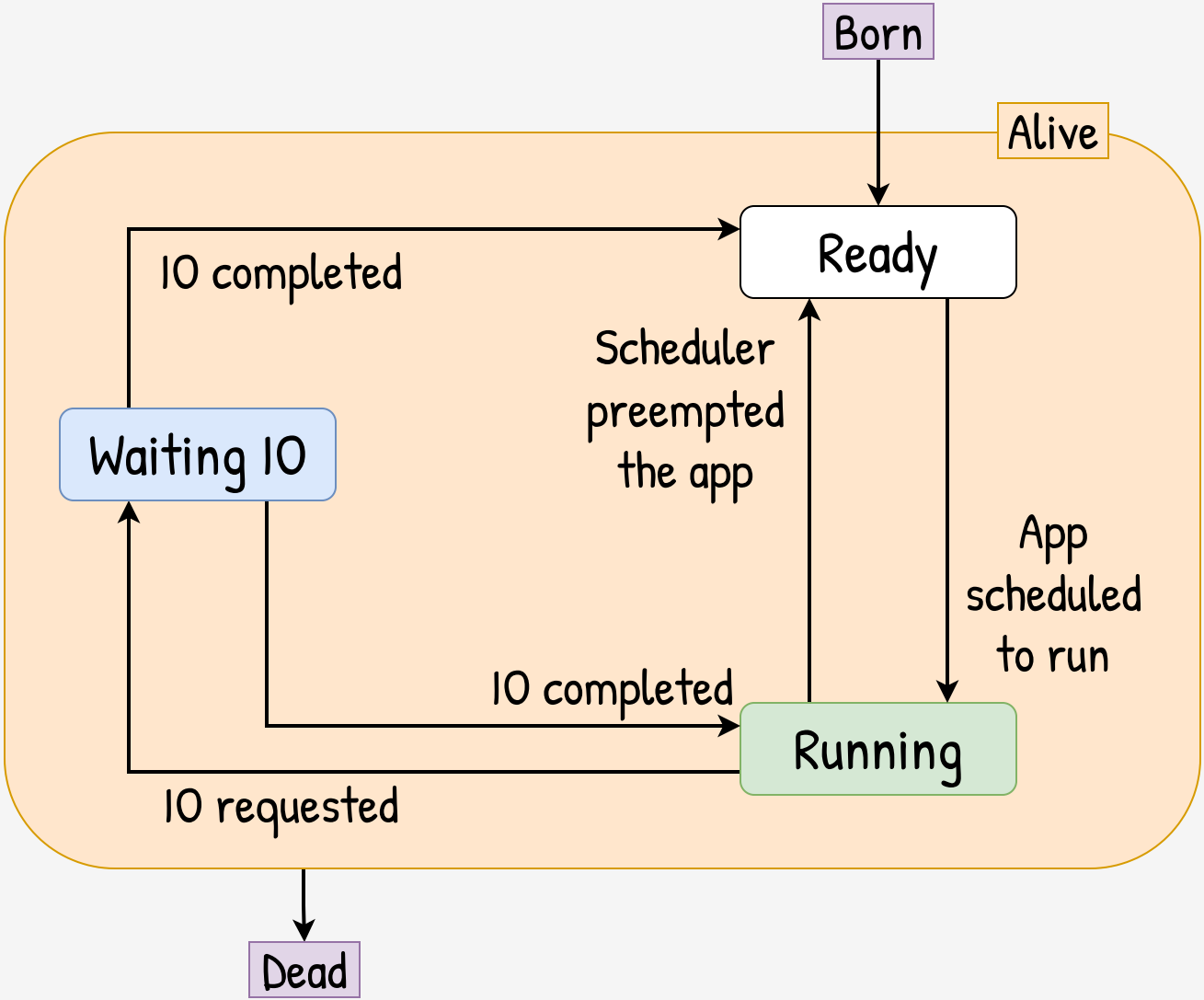



CPU's perspective
x% usage
$(x \approx 100)$
Scheduler ***switches*** app 10 waiting for one ready to run
x% usage
$(x < 100)$

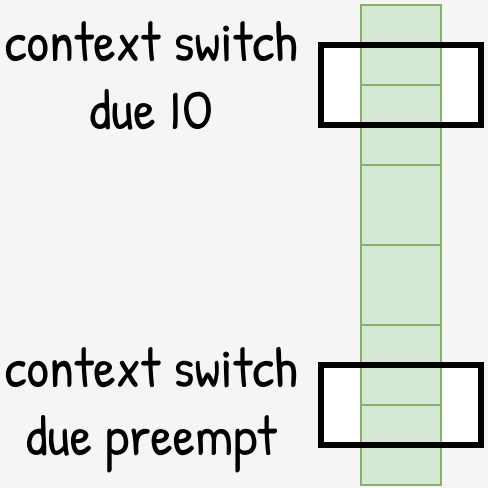

CPU's perspective
x% usage
$(x \approx 100)$
After IO completion, the app may ***not*** resume if other app is running
x% usage
$(x < 100)$

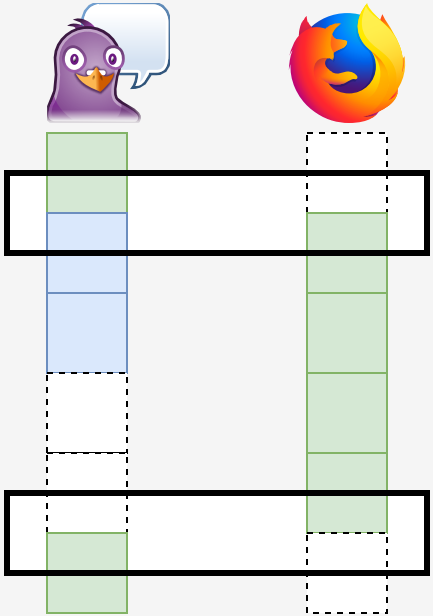

CPU's perspective
App neither running nor IO waiting, it is ready to run but waiting for CPU
x% usage
$(x \approx 100)$
x% usage
$(x < 100)$

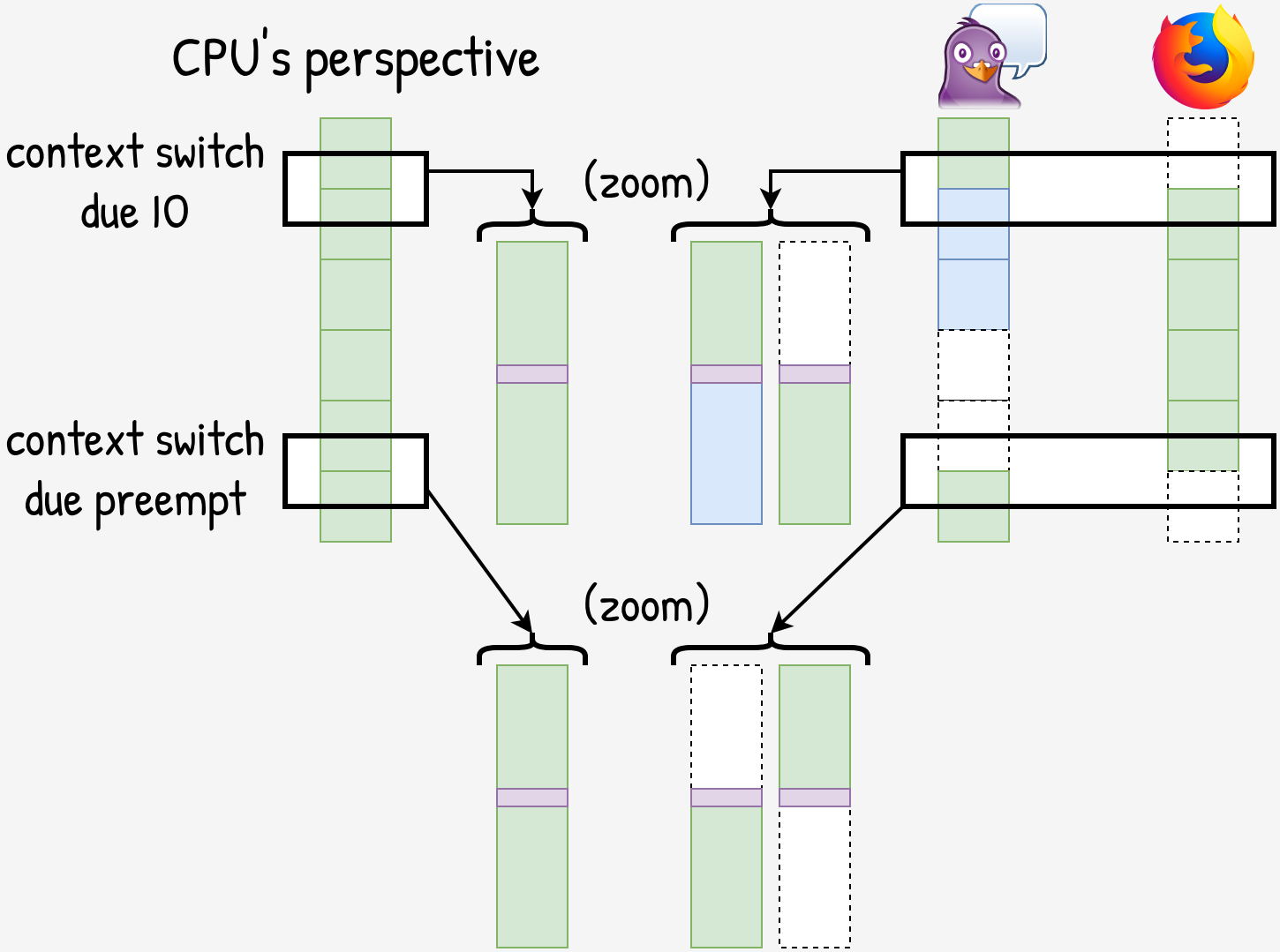

CPU's perspective
x% usage
$(x \approx 100)$
Schedule switches app after a while to be nice with others (*preempts* the CPU)
x% usage
$(x < 100)$


Born
Alive
Ready
IO completed
Scheduler preempted the app
Waiting IO
App scheduled to run
IO completed
Running
IO requested
Dead

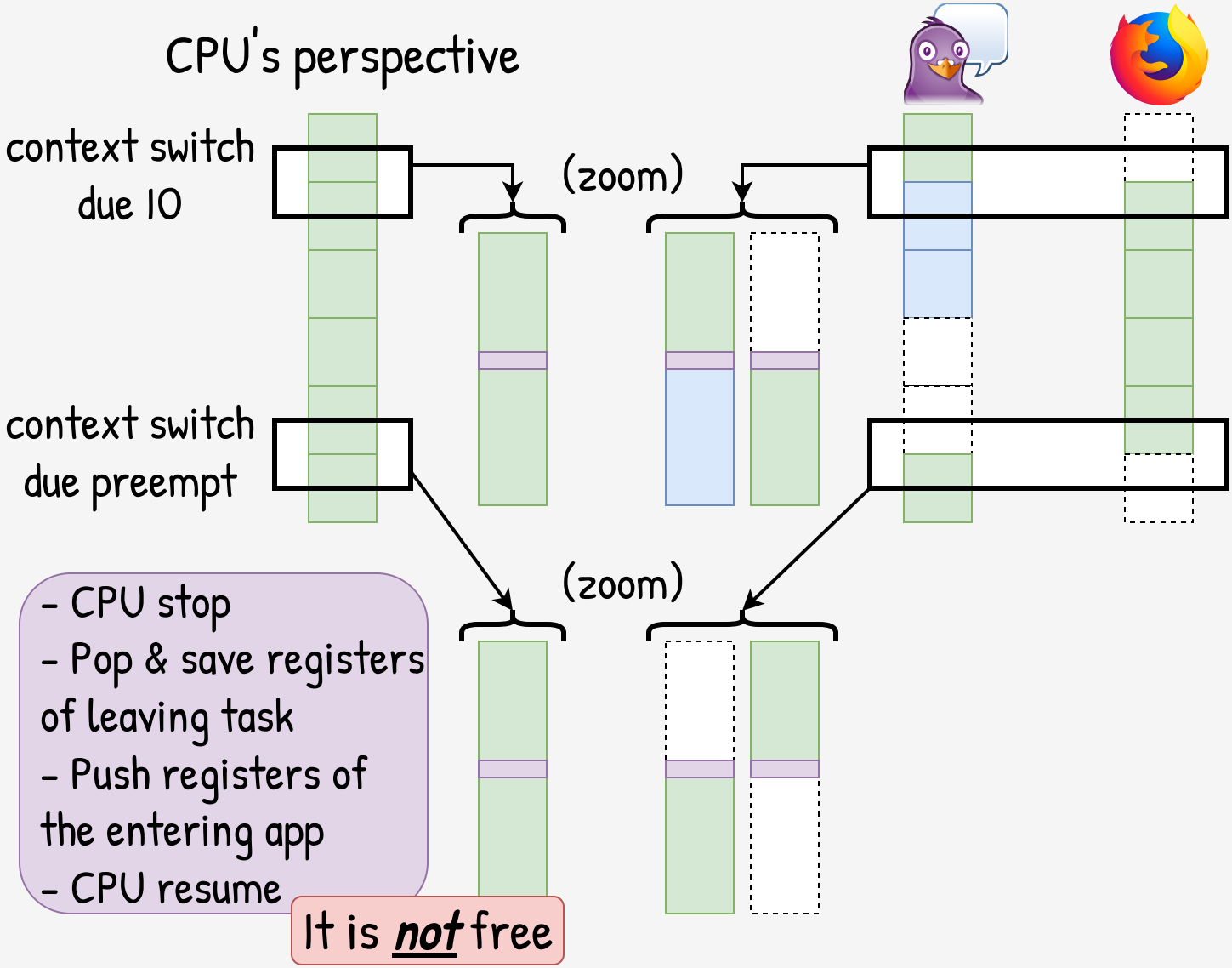
CPU's perspective

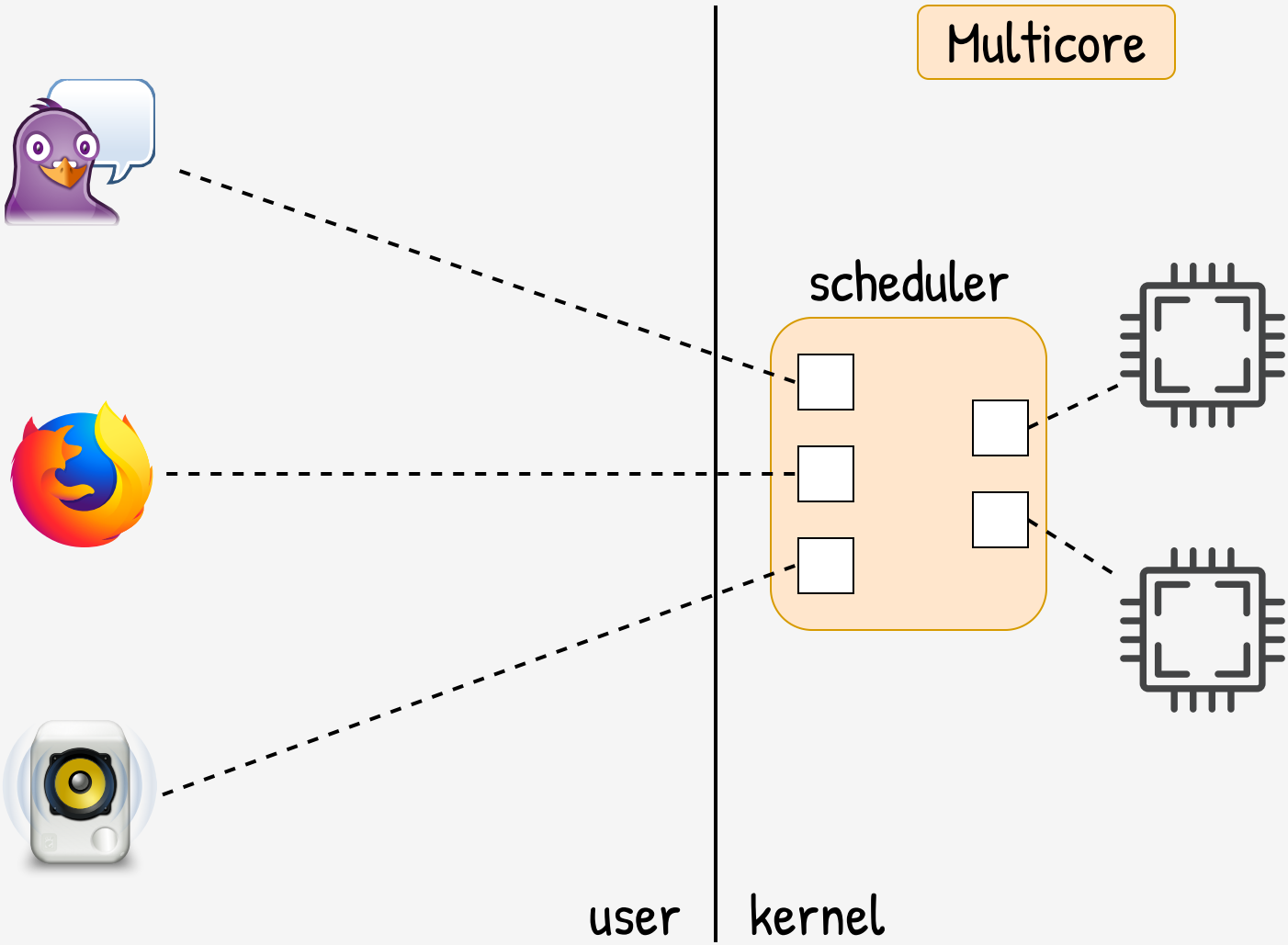


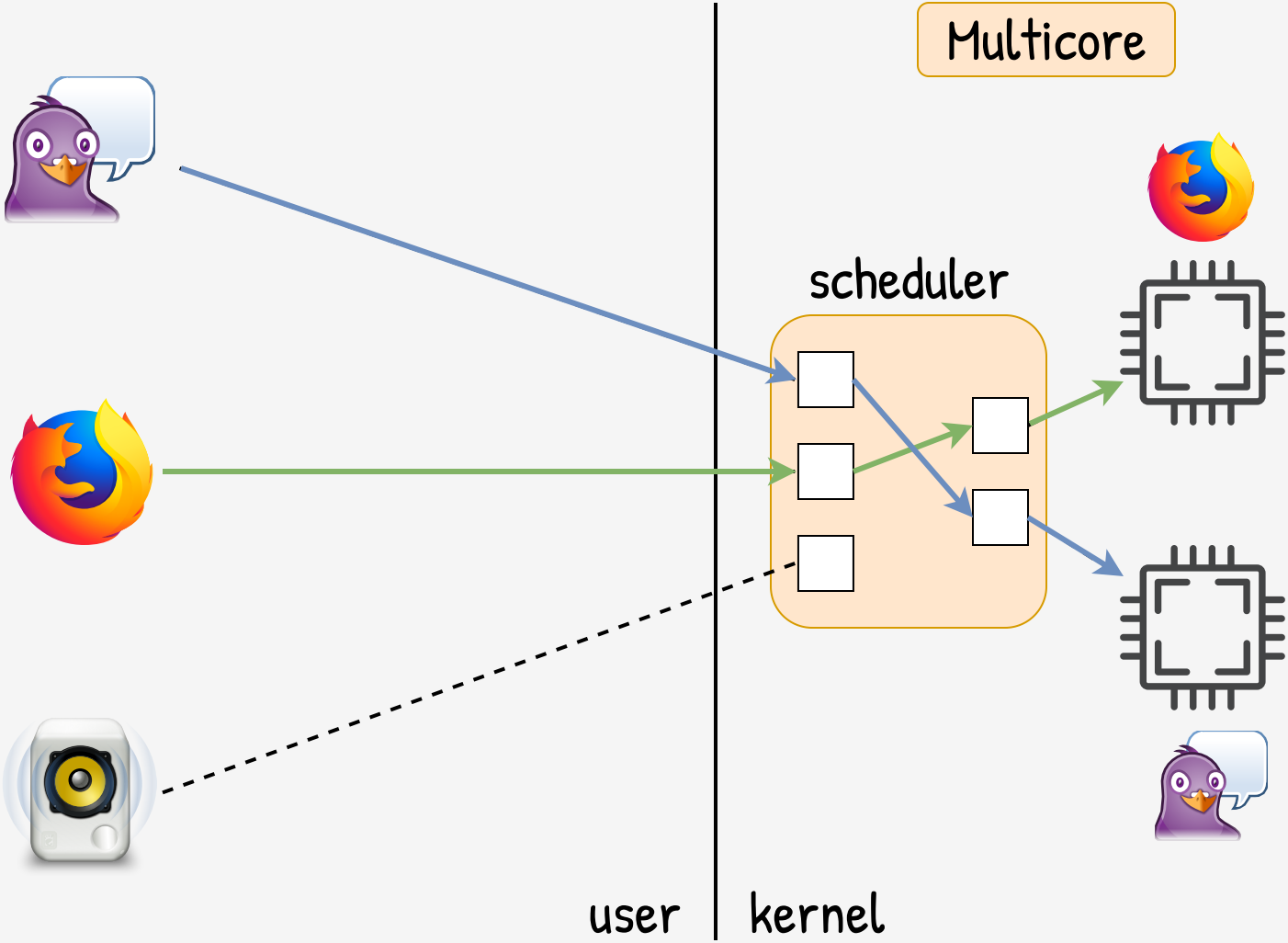



CPU's perspective
context switch
due IO
(zoom)
context switch
due preempt
(zoom)

CPU's perspective




Multicore
scheduler
user
kernel


Multicore
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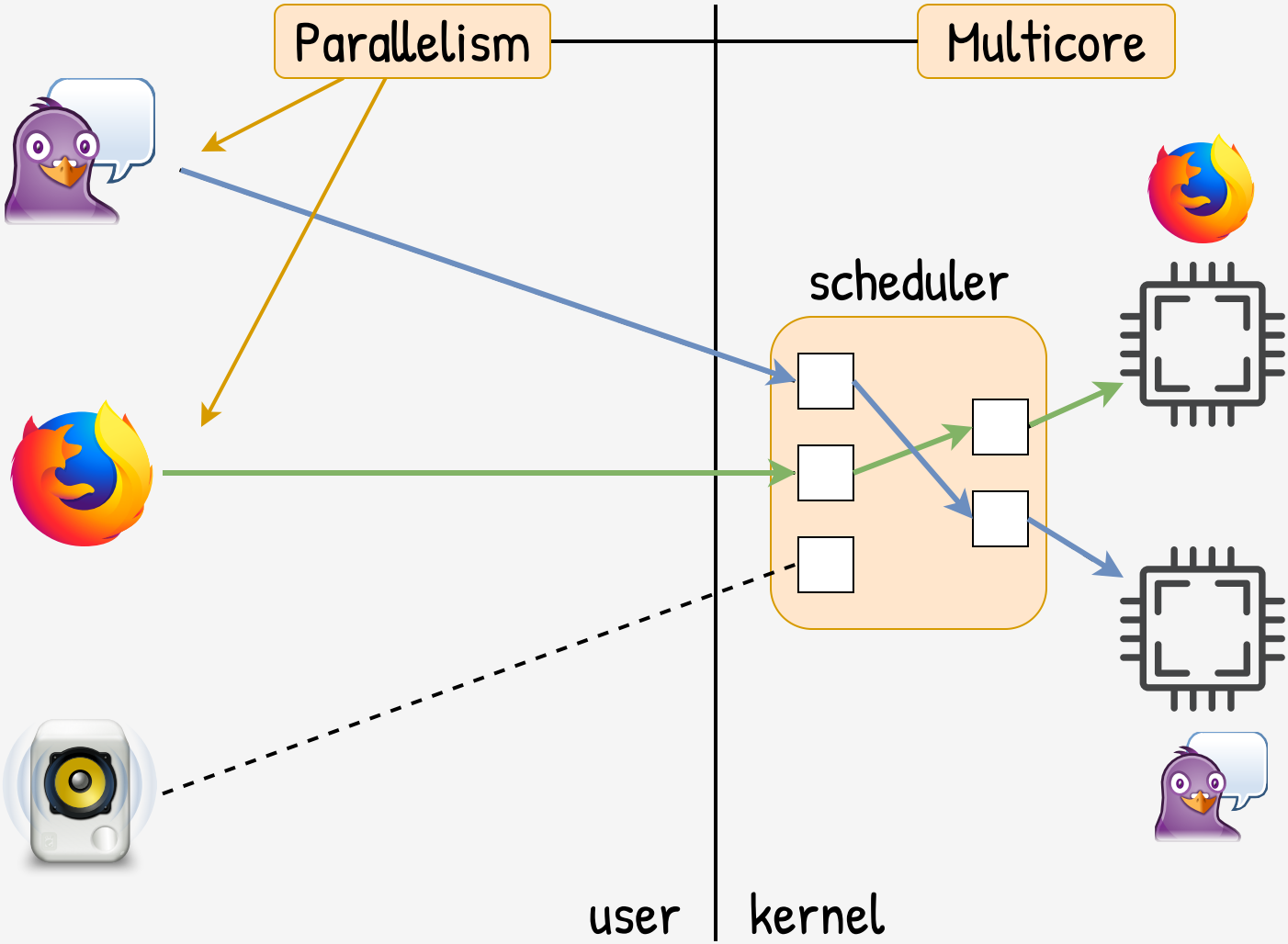
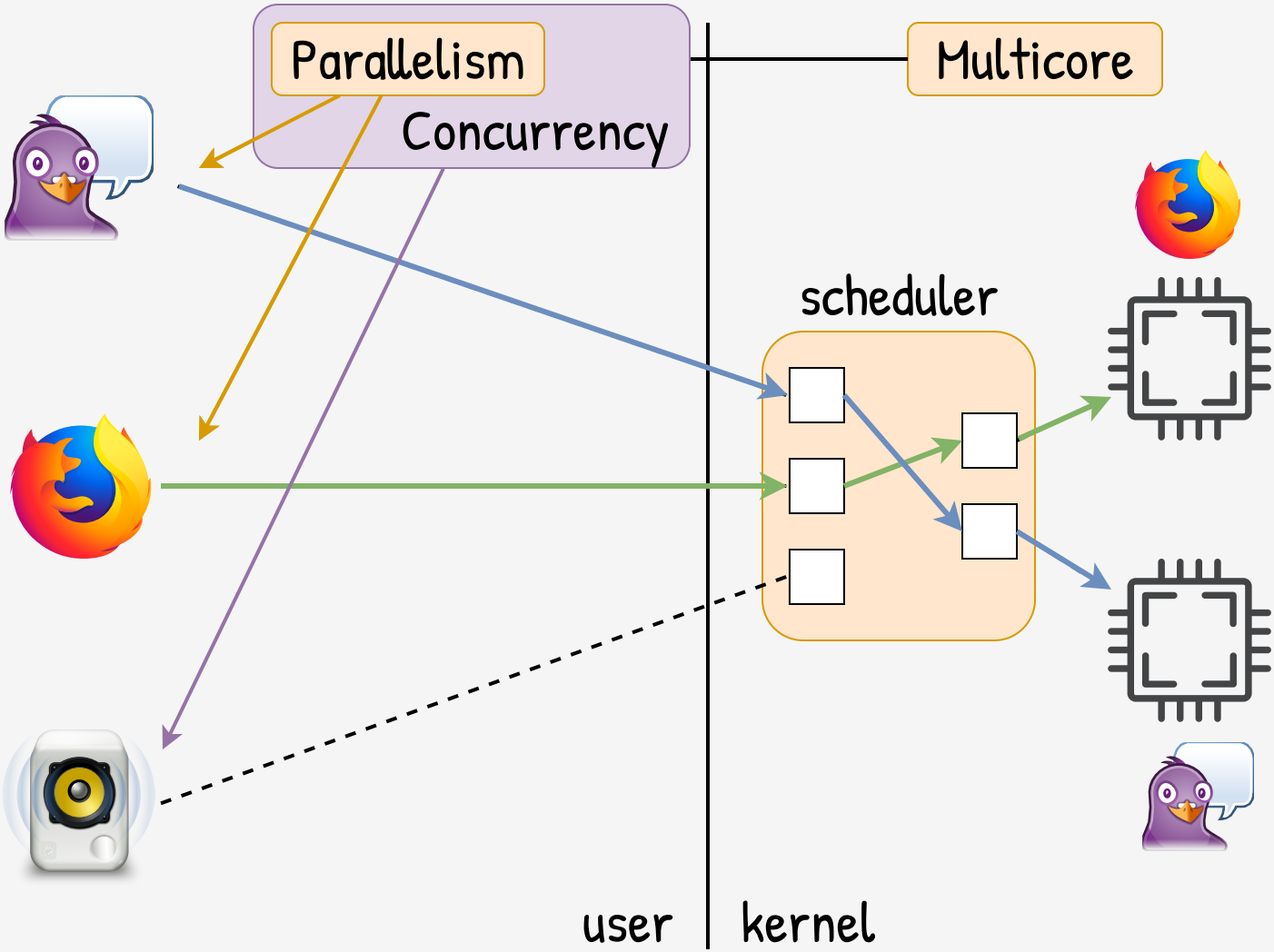
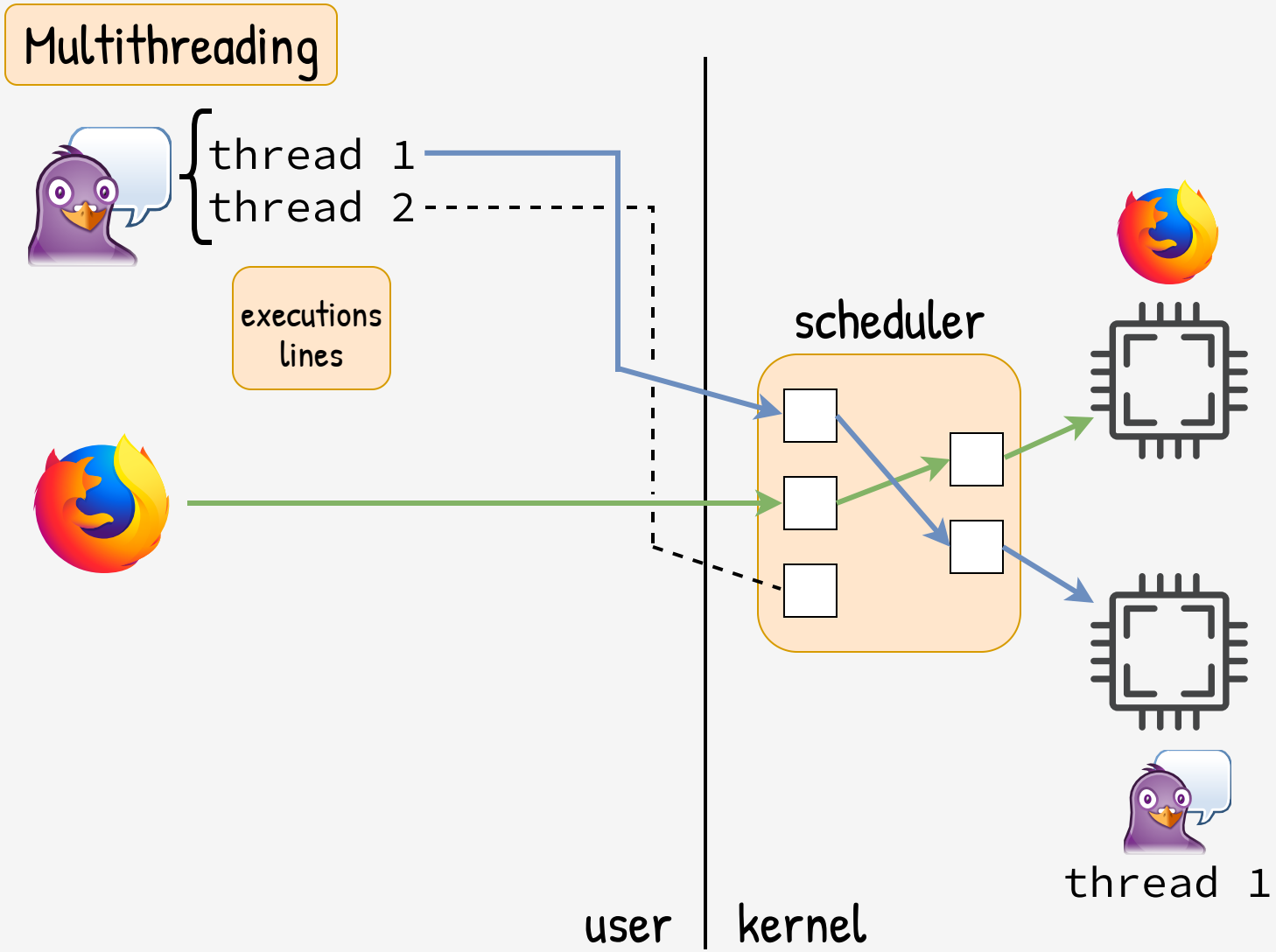
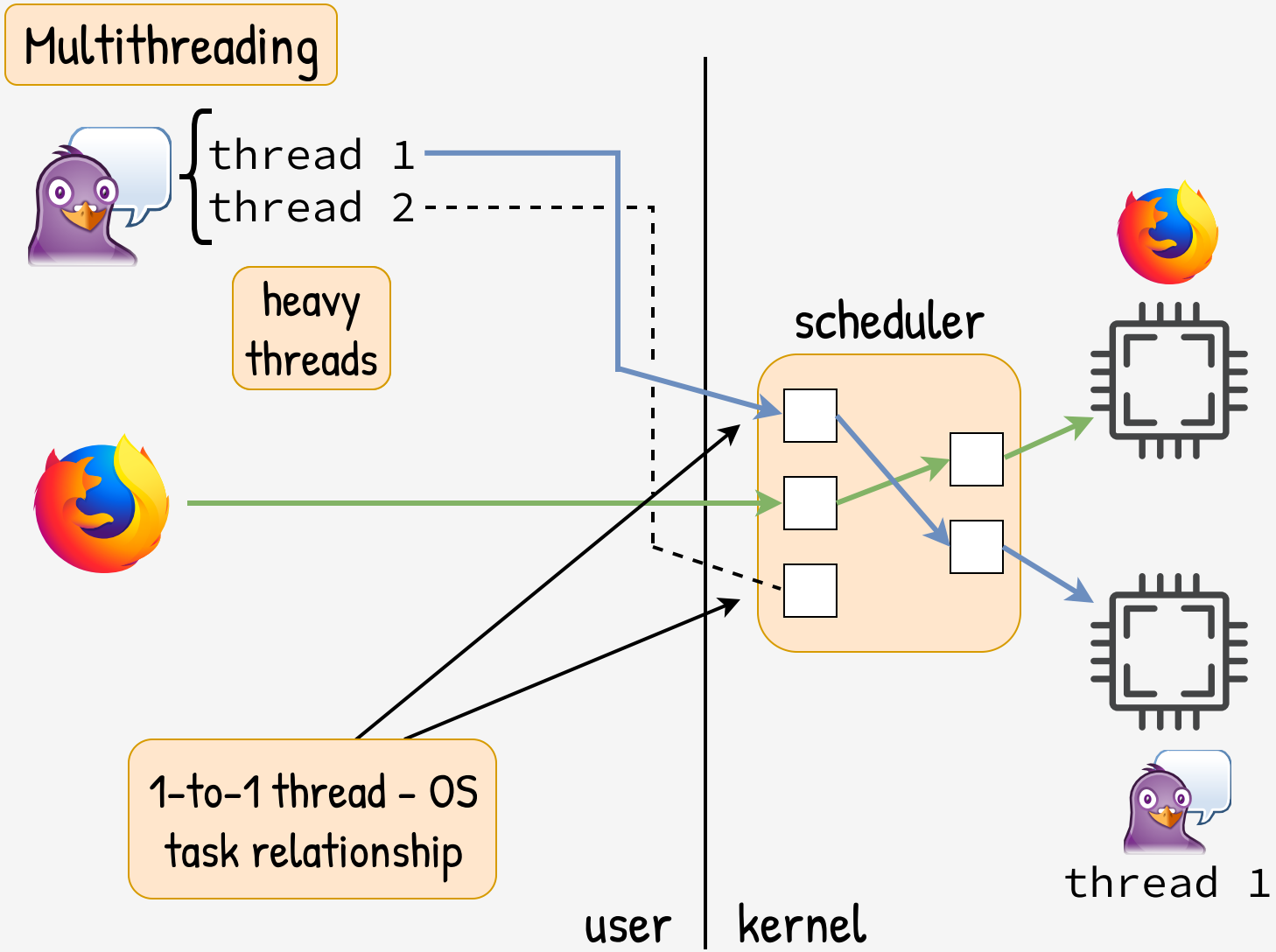
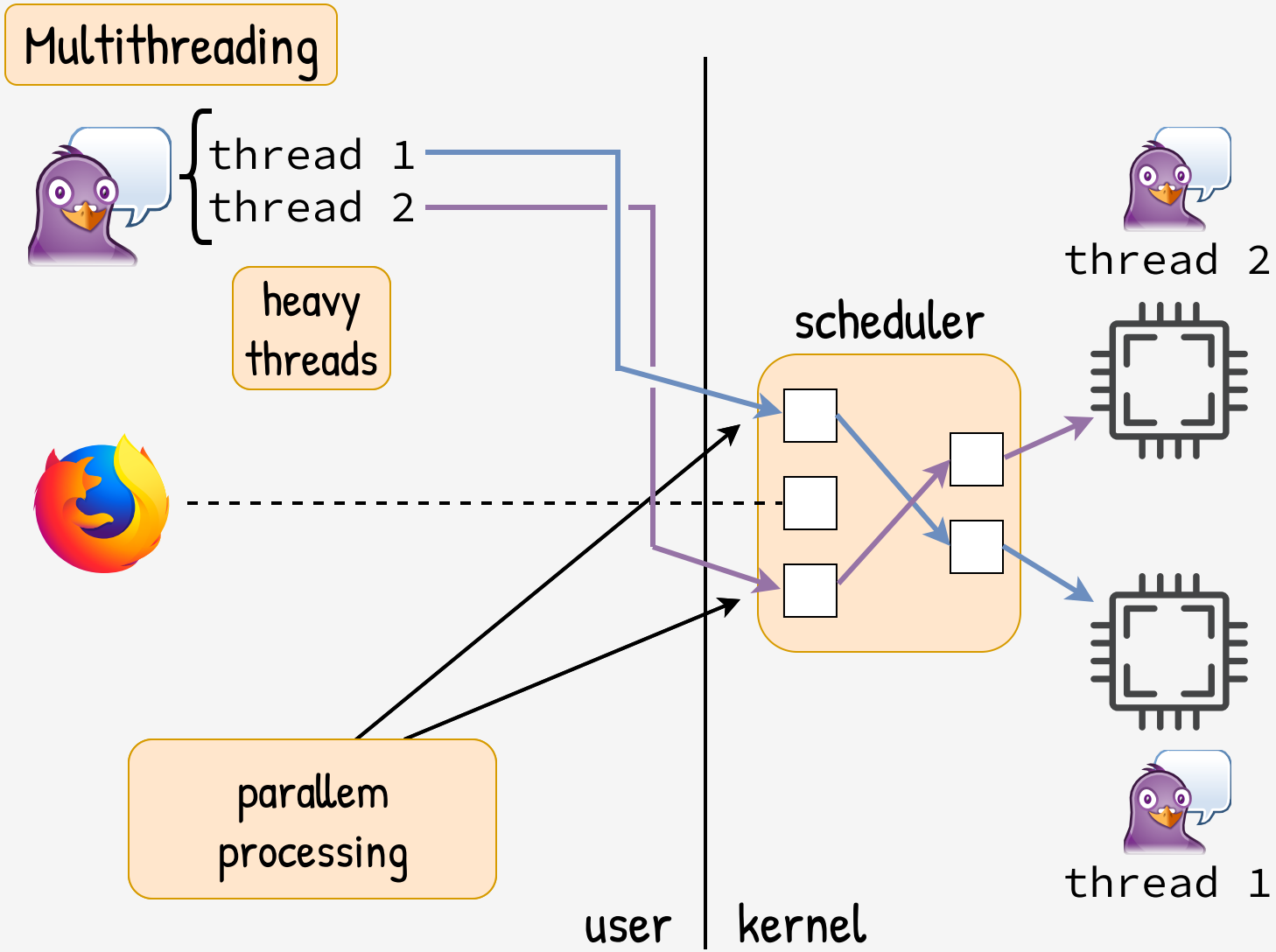
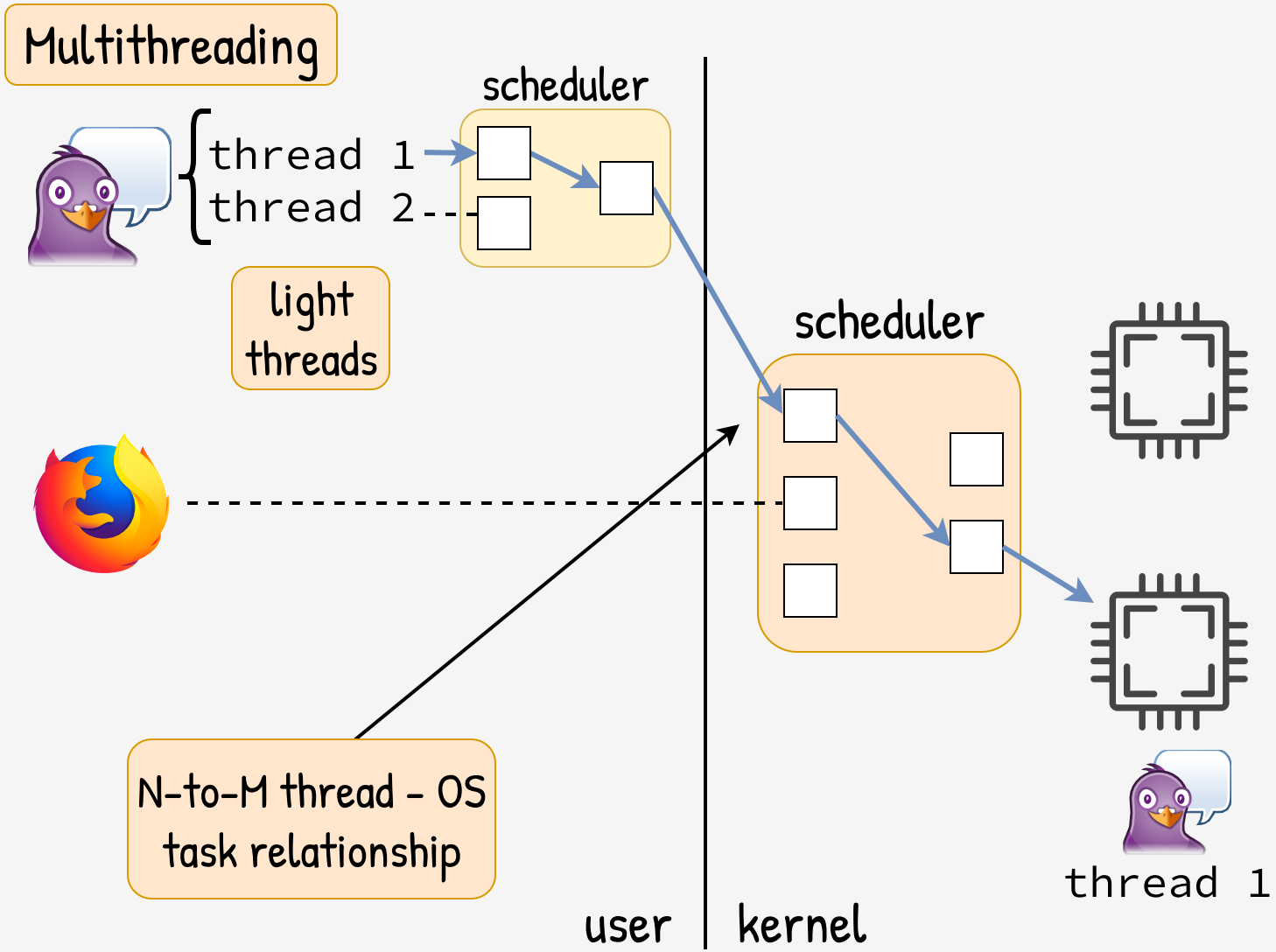

Parallelism
Multicore
scheduler
user
kernel


Parallelism
Concurrency
Multicore
scheduler
user
kernel


Multithreading
thread 1
thread 2
executions lines
scheduler
thread 1
user
kernel


Multithreading
thread 1
thread 2
heavy threads
scheduler
1-to-1 thread - OS task relationship
user
kernel
thread 1


Multithreading
thread 1
thread 2
heavy threads
scheduler
thread 2
parallem processing
thread 1
user
kernel


Multithreading
scheduler
thread 1
thread 2
light threads
scheduler
N-to-M thread - OS task relationship
thread 1
user
kernel

Takeaways

Single thread process?

Multi-processes?

Single process, multi-threads?

Heavy or light threads?

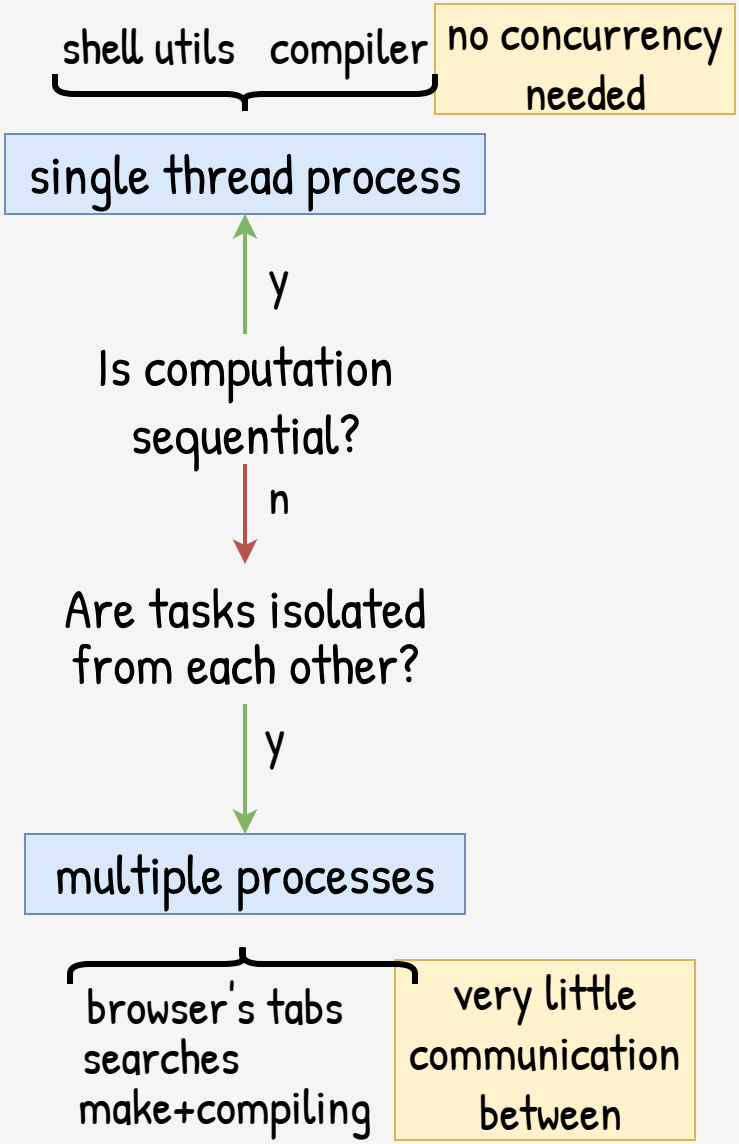
shell utils compiler

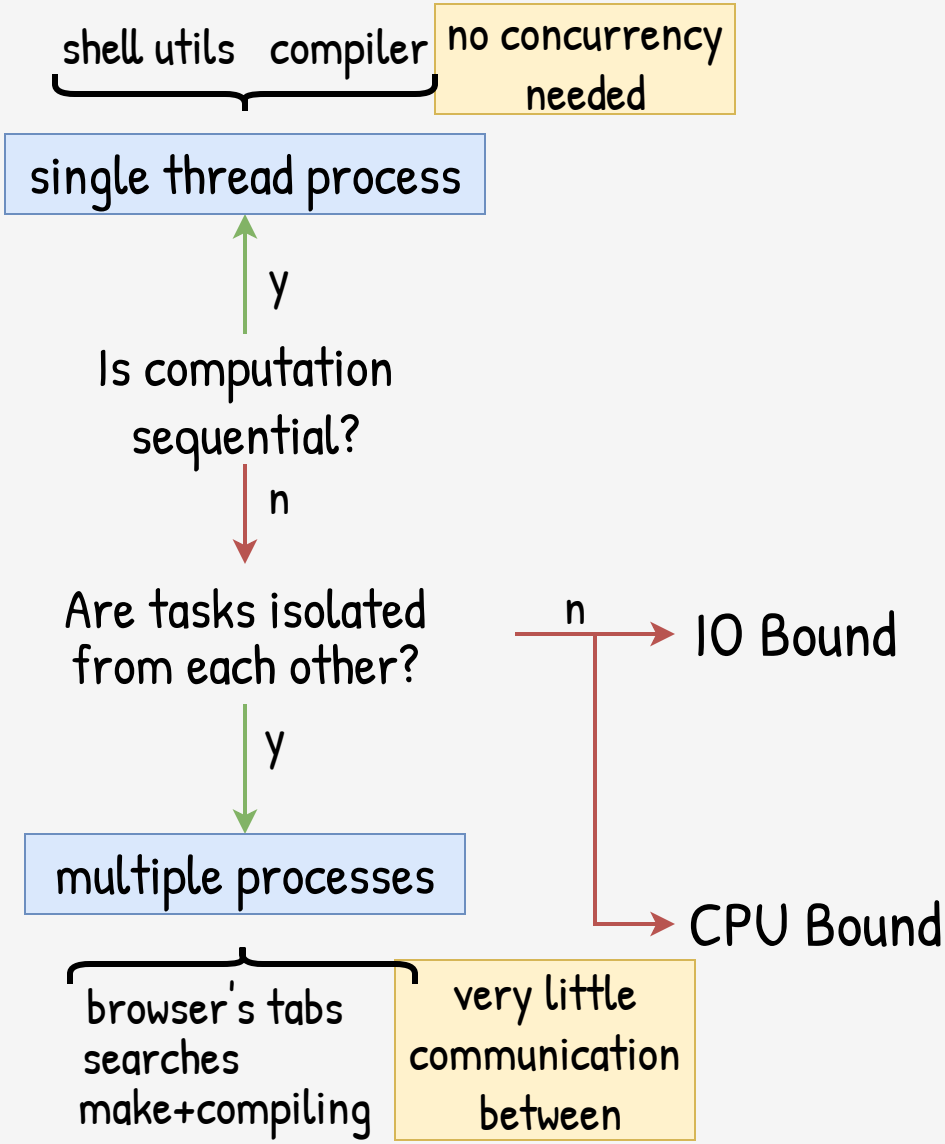
no concurrency needed

single thread process

y

Is computation sequential?


shell utils
compiler
no concurrency needed
single thread process
y
Is computation sequential?
n
Are tasks isolated from each other?
y
multiple processes
browser's tabs
searches
make+compiling
very little communication between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IO Bound
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multiple processes
CPU Bound
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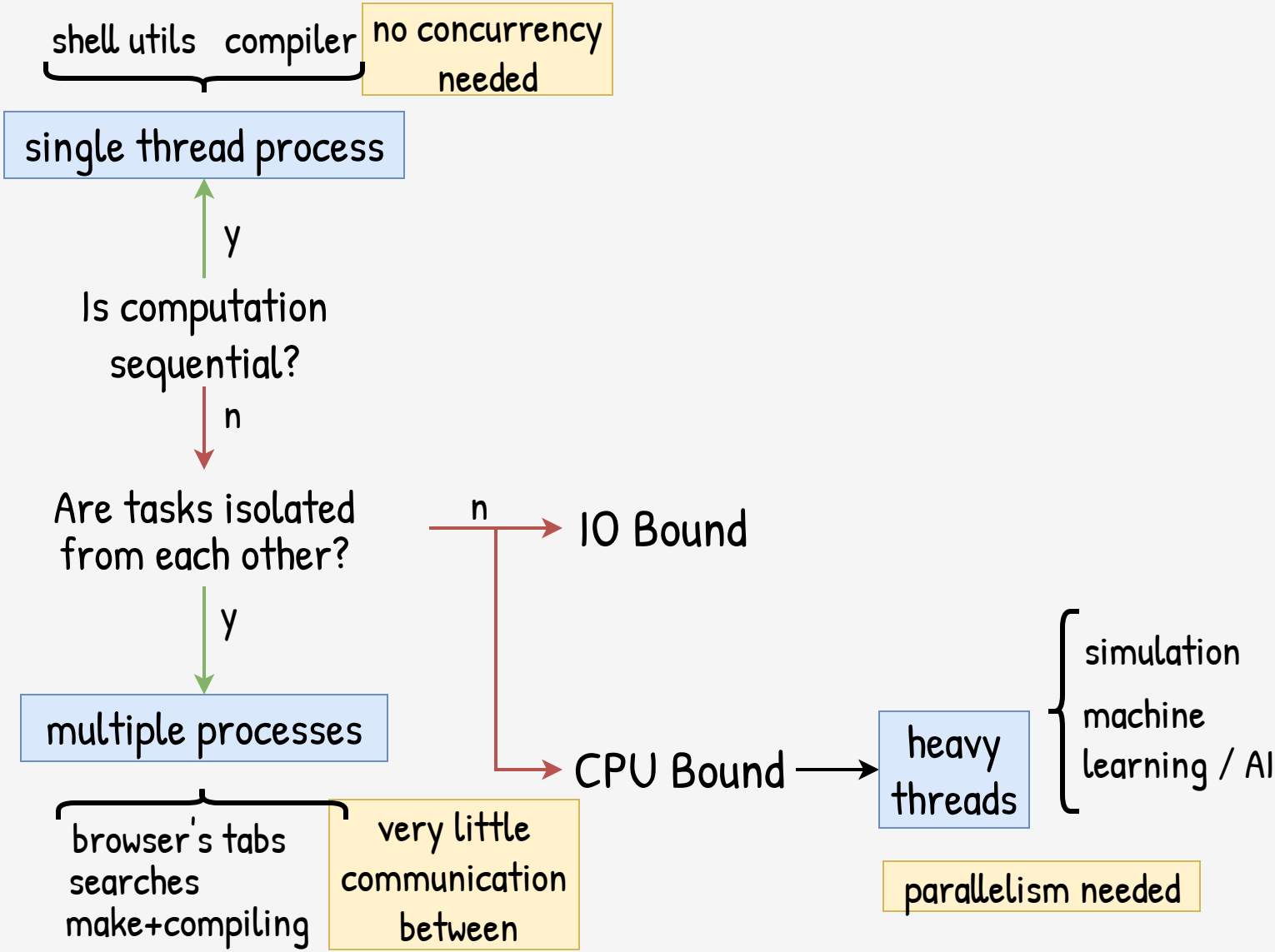
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CPU Bound
heavy threads
simulation
machine learning / AI
browser's tabs
searches
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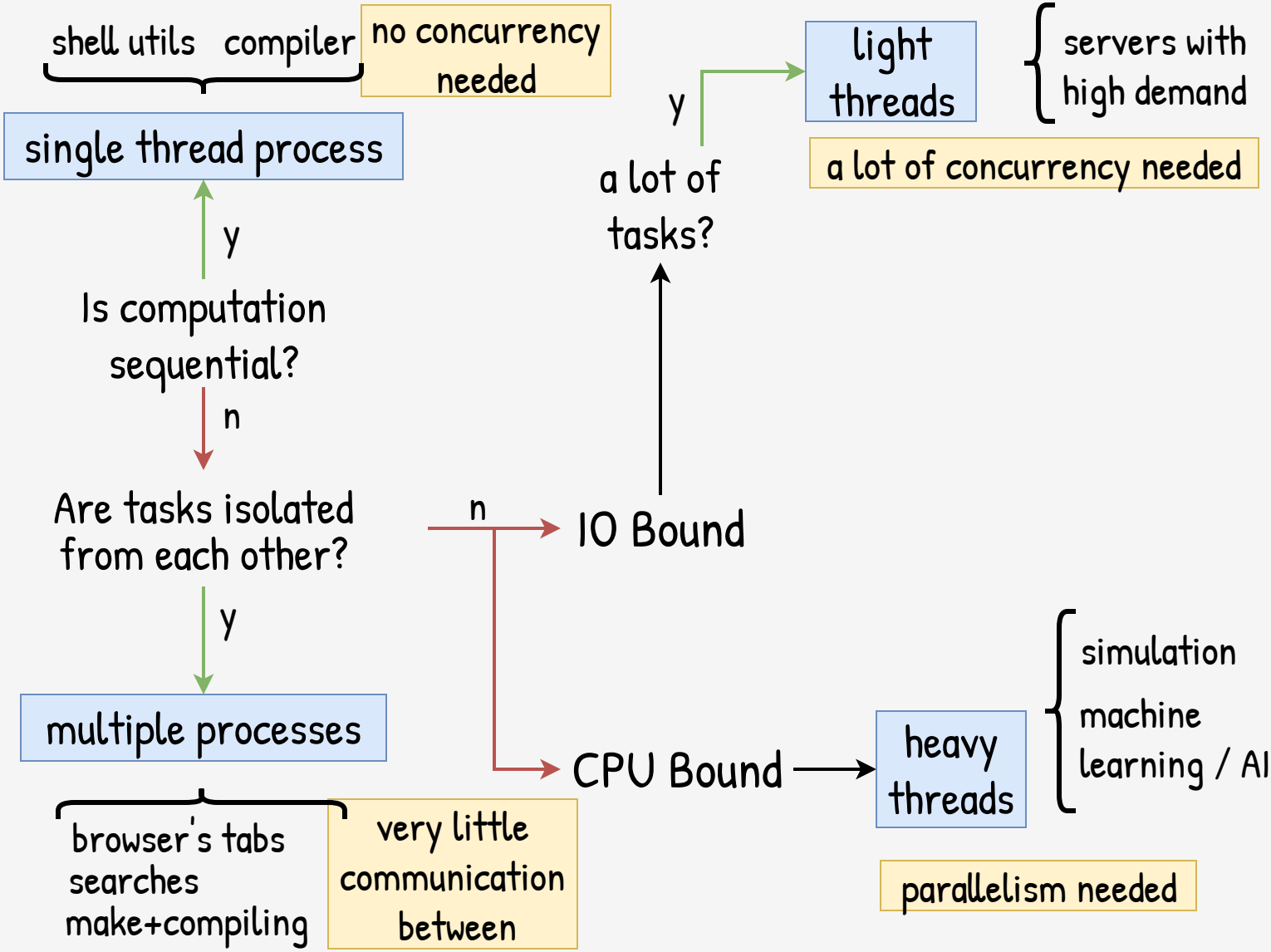
very little communication between
parallelism needed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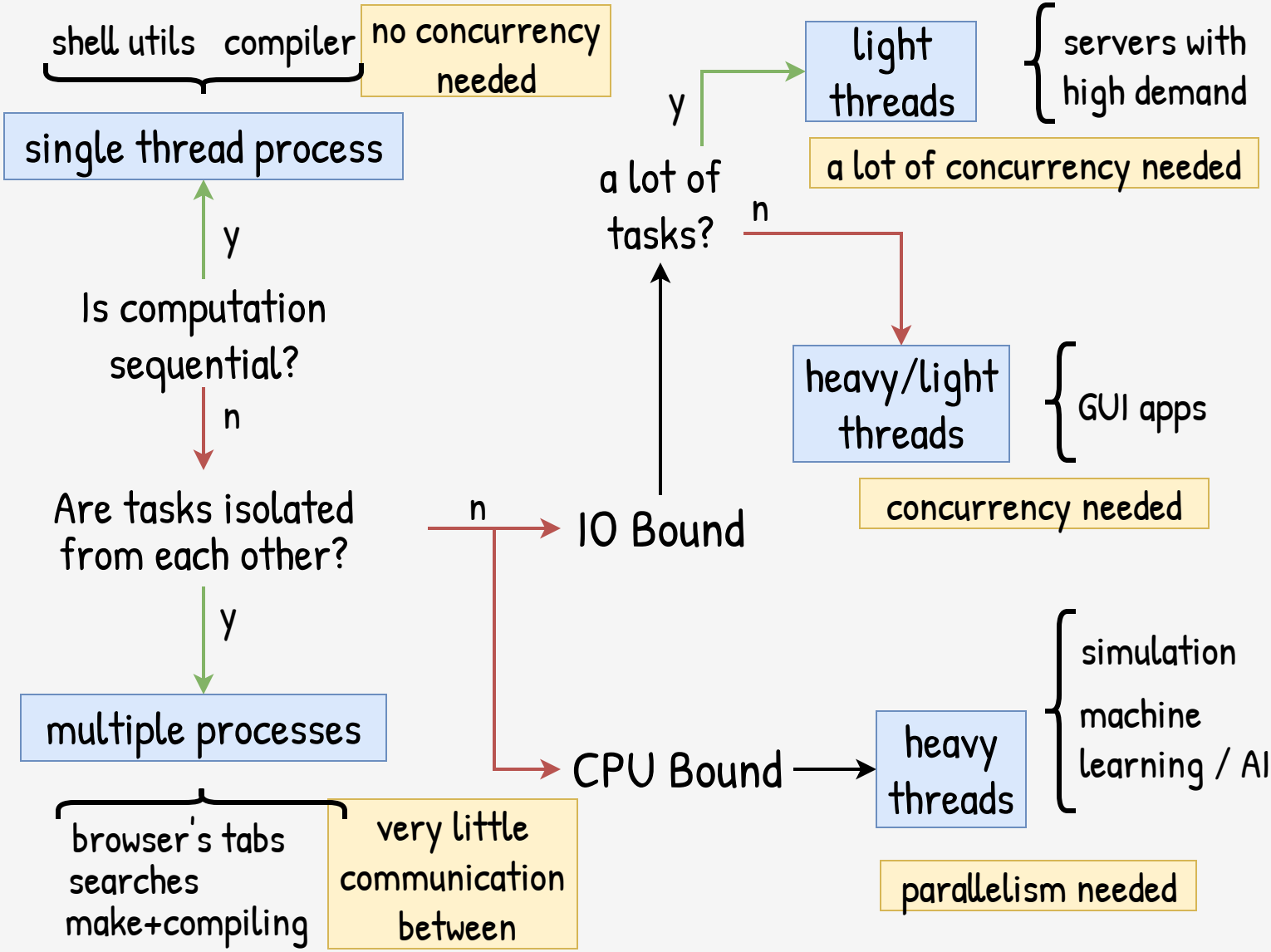
make+compiling
very little communication between
n
IO Bound
y
a lot of tasks?
light threads
servers with high demand
a lot of concurrency needed
CPU Bound
heavy threads
simulation
machine learning / AI
parallelism needed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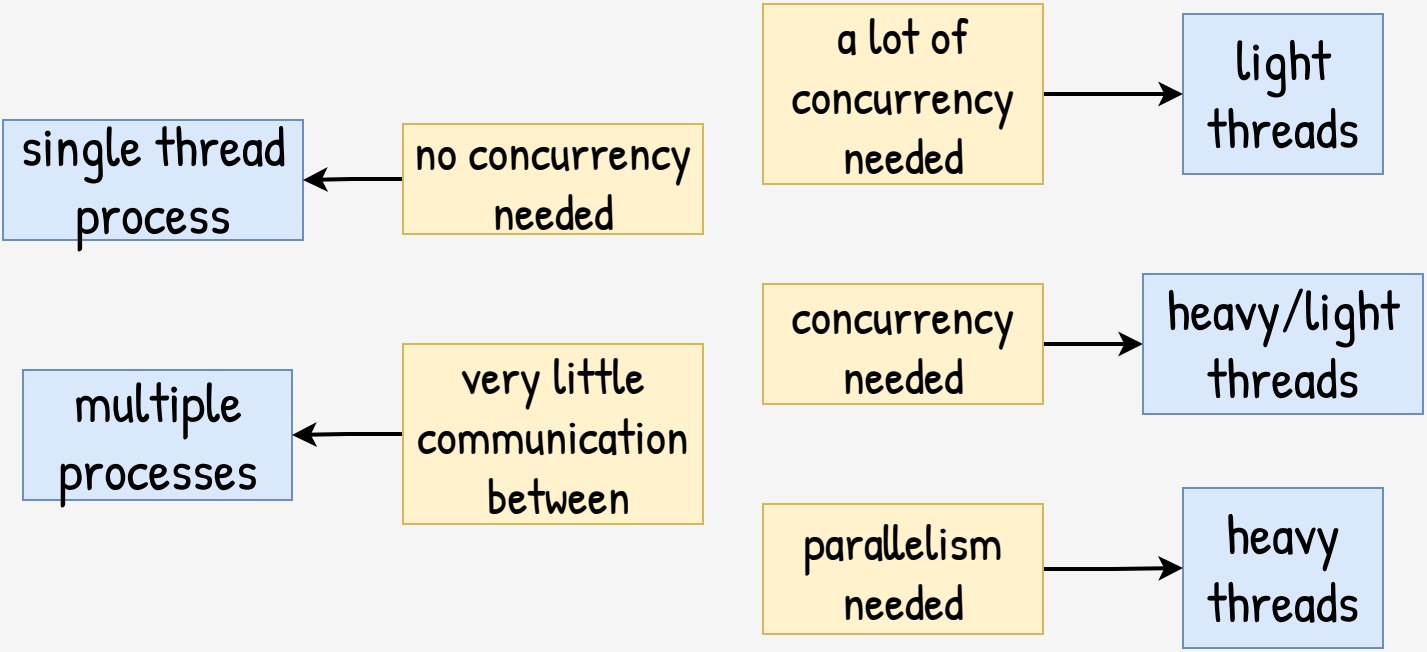
a lot of tasks?
y
light threads
servers with high demand
a lot of concurrency needed
n
heavy/light threads
GUI apps
concurrency needed
CPU Bound
heavy threads
simulation
machine learning / AI
parallelism needed

Takeaways


single thread process
no concurrency needed
multiple processes
very little communication between
a lot of concurrency needed
light threads
concurrency needed
heavy/light threads
parallelism needed
heavy threads